Module Specification

Distributed and Enterprise Software Development

Version: 2026-27, v5.0, Approved

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Part 1: Information

Module title: Distributed and Enterprise Software Development

Module code: UFCFTR-30-3

Level: Level 6

For implementation from: 2026-27

UWE credit rating: 30

ECTS credit rating: 15

College: College of Arts, Technology and Environment

School: CATE School of Computing and Creative Technologies

Partner institutions: None

Field: Computer Science and Creative Technologies

Module type: Module

Pre-requisites: Advanced Software Development 2025-26, Principles of Programming 2024-25, Systems Development Group Project 2025-26

Excluded combinations: None

Co-requisites: None

Continuing professional development: No

Professional, statutory or regulatory body requirements: None

Part 2: Description

Overview: This module consolidates the previously attained programming and systems development knowledge in years 1 & 2 and prepares students for the challenges of solving and implementing solutions for complex organisations such as enterprises. These are decentralised, networked multilevel systems that produce and process large volumes of data. Being able to assess the suitability, select the right

tools and techniques and implement enterprise wide systems is an important set of knowledge and valuable skill set for a systems developer.

Features: Not applicable

Educational aims: This module aims to introduce enterprise level software development using contemporary distributed technologies.
It also aims at developing students understanding of enterprise system development and to contribute to their understanding of legal, ethical, social and professional aspects.

Outline syllabus: The module initially teaches the concepts and paradigms of enterprise systems including distributedness, virtualisation, synchronisation, scalability, integrity, latency, fault-tolerance and security to implement small to large-scale software applications. Here, students will learn various architectural patterns for enterprise systems and study n-tier architecture, service oriented and microservices architectures in detail. Theory and concepts of components, interfaces and services will be taught by using a suitable modelling notation. Students will also learn to design and implement trust and security concepts in implementing secure large scale software systems.

Students will also learn enterprise system development and associated legal, ethical, social and professional aspects. This will cover understanding about enterprises of planetary scale and associated complexity. Software systems by acquisition, integration, configuration and customisation, and subsequent interoperability challenges will be covered. The economics of various software acquisition strategies, from DIY to components-of-the-shelf (COTS), to software packages, to service-based capabilities in the Cloud and Edge computing will be covered. Use of virtualisation and containerisation will be taught to understand various scalability, performance, storage, process and deployment related issues associated with enterprise scale software development in a cloud/edge environment. The significance, capabilities and examples of contemporary technologies such as blockchain or distributed ledgers and cloud and edge computing will be explored in the context of enterprise system development. Students will test these technologies and find out how these technologies impact enterprise software development.

They will learn and apply an agile approach for developing enterprise software by using an enterprise scale framework and gain clear understanding of complexity and challenges associated to stakeholder and requirements analysis, design, implementation, testing and maintenance processes.

Part 3: Teaching and learning methods

Teaching and learning methods: The module will be delivered via a combination of lectorial/workshop and lab sessions, with face-to-face and online help provided by tutors. Where relevant, employers will be invited to give a guest lecture on their company's development practices.

Lectures will focus on providing basic concepts and introduction to lab sessions and independent learning. Lab sessions will focus on allowing the students to apply the concepts learned in the lectures to various problems and contexts. Online resources such as UWE e-library and LinkedIn learning are also available.

Students have the opportunity to practise an agile software development approach by working collaboratively in groups and receiving continuous formative feedback from the module team through a variety of stand-up meetings and sprint reviews.

Module Learning outcomes: On successful completion of this module students will achieve the following learning outcomes.

MO1 Describe and evaluate the concepts and paradigms of modern enterprise systems with particular focus on architectures, components, interfaces and services.

MO2 Evaluate the legal, ethical, social and professional implications in development life-cycle of large scale software systems.

MO3 Apply a current industrial project management approach when undertaking a large-scale software development project.

MO4 Identify security issues in distributed or enterprise level software systems in order to implement preventive measures.

Hours to be allocated: 300

Contact hours:

Independent study/self-guided study = 228 hours

Staff-guided learning = 72 hours

Reading list: The reading list for this module can be accessed at readinglists.uwe.ac.uk via the following link from https://rl.talis.com/3/uwe/lists/874C62FC-4042-00DE-7C1C-CD496104F65F.html

Part 4: Assessment

Assessment strategy: The assessment has been designed to ensure that students' understanding and skills are developed incrementally and the assessment strategy provides continual formative verbal feedback opportunities and allows students to develop their skills with the materials being presented in the lectures and laboratory sessions. The group-based working also provides numerous peer-learning opportunities.

The assessment will comprise two tasks: (1) a group based report and (2) a group project developed and demonstrated.

The report tests students understanding of theoretical concepts of enterprise system development through a reflective and an evaluative report written on how analysis, modelling and architectural aspects of a software development project is done following professional skills developed through lecture and practical sessions. Each student will have opportunities to demonstrate the practical knowledge and transferable skills developed so far. Students are expected to reflect on legal, ethical, social and professional aspects of the software development project, which is expected to let students prepare for the group project.

The project will allow students to practice team work via a group project to prepare for professional life. It will let students implement the practical knowledge and

transferrable skills following the good practices to contribute to group activities. There will be a group mark, and an individual contribution weight calculated from peer assessment data as well as tutor observations. The individual mark will be derived from these two components (group mark and individual weight) and will be supported with individual performance in Q/As during the demonstration. The tutors' observation will help adjustment the marks if evidence of significant unbalanced contributions from the group members is sensed. Each group will submit project code, and be expected to demo their completed project illustrating both group and individual programming skills. Students will get verbal and written feedback.

Resit assessment will match the first sit and deliverable(s) will be scaled appropriately to group size and task complexity.

Assessment tasks:

Project (First Sit)
Description: Group project with demonstration (30 Minutes)
Weighting: 60 %
Final assessment: No
Group work: Yes
Learning outcomes tested: MO2, MO3, MO4

Report (First Sit)
Description: A group reflection report on small project and professional practices (1000-3000 words)
Weighting: 40 %
Final assessment: Yes
Group work: Yes
Learning outcomes tested: MO1, MO2

Project (Resit)
Description: Group project with demonstration (30 Minutes)
Weighting: 60 %
Final assessment: No

Group work: Yes
Learning outcomes tested: MO2, MO3, MO4

Report (Resit)
Description: A group reflection report on small project and professional practices (1000-3000 words)
Weighting: 40 %
Final assessment: Yes
Group work: Yes
Learning outcomes tested: MO1, MO2

Part 5: Contributes towards

This module contributes towards the following programmes of study:

Computer Science {Foundation} [Frenchay] BSc (Hons) 2023-24

Computer Science [Phenikaa] BSc (Hons) 2024-25

Computer Science (Artificial Intelligence) [NepalBrit] BSc (Hons) 2024-25

Computer Science [Frenchay] BSc (Hons) 2024-25

Computer Science [Villa] BSc (Hons) 2024-25

Computer Science {Foundation} [GCET] BSc (Hons) 2022-23

Computer Science (Artificial Intelligence) {Foundation} [GCET] BSc (Hons) 2022-23

Computer Science (Smart Devices) {Foundation} [GCET] BSc (Hons) 2022-23

Computer Science (Artificial Intelligence) {Foundation} [GCET] BSc (Hons) 2023-24

Computer Science {Foundation} [GCET] BSc (Hons) 2023-24

Computer Science (Smart Devices) {Foundation} [GCET] BSc (Hons) 2023-24

Computer Science [Frenchay] BSc (Hons) 2024-25

Computer Science {Dual} [Taylors] BSc (Hons) 2024-25

Computer Science {Dual} [Taylors] BSc (Hons) 2024-25

Computer Science {Foundation} [Frenchay] BSc (Hons) 2022-23

Computer Science [Frenchay] BSc (Hons) 2023-24

Computer Science (Smart Devices) {Foundation} [GCET] BSc (Hons) 2023-24

Computer Science (Artificial Intelligence) {Foundation} [GCET] BSc (Hons) 2023-24

Computer Science {Foundation} [GCET] BSc (Hons) 2023-24